CS 230 – REQUIREMENTS ENGINEERING

ASSIGNMENT 1

SEMESTER 1, 2025

STUDENT ADMINISTRATION SYSTEM

FEASIBILITY REPORT

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1.0 Scope and Viability

1.1 Scope

1.1.1 Project Name

Student Administration System (SAS)

1.1.2 Project Requirement

The goal of this project is to develop a convenient modern unified digital platform to manage all student information, enrolment, records, attendance and extracurricular activities. The system should ensure data security and compliance of student privacy policy

1.1.3 End Users

1. Students
2. School Administrators
3. Faculty Members
4. Legal Guardian

1.1.4 Modules to Be Included in This Project

According to the project requirement, this website will be divided into multiple modules as follow

1. A repository to store student information
2. A report card system and tracker of academic achievements
3. Tracker and management module of discipline and attendance
4. Course timetable enquiry and course scheduling
5. Tracker of extracurricular activities
6. Platform for communication between parents and school .
7. Management module for administrators
8. Library and learning resources repository
9. Data security and privacy system

1.1.5 Development Process

1. Requirements gathering
2. System design
3. Development
4. Testing
5. Deployment/Implementation

1.2 Feasibility

1.2.1 Limitations

There are limitations of 4 aspects encountered with this project.

1. Economic Limitation

The expense amount of the project, including costs of software, hardware, training in the use of the system, deployment and maintenance will affect the feasibility. The project will fail once the overall expense is out of control.
2. Technical Limitation
Requirements for developer competence depends on the software type, user devices, user experience, privacy and security, scalability, testing will be limitations. Any of these requirements is vital for the project.
3. Resource Limitation
How to allocate team resources effectively? How stable is the Internet infrastructure? These resource limitations will decide the quality and the availability of the system.
4. Time Limitation
Time expense of the overall project will determine if the project will be finished within a reasonable time period. Once time schedule is over control, the project will fail.

1.2.2 Feasibility

Those limitations are able to be managed.
1. Economic
Open-source tools, cloud hosting, free workshops for training, pilot deployment approach and annual system update will minimize the expense, which makes the overall expense under control.
2. Technical
The project will be able to meet requirements by using open-source tools, modern frameworks, proven and accredited technologies, and scalable cloud architecture together with the user-centred design methodology
3. Resource
Clear task assignment and role differentiation enable rational allocation of team resources. A small amount of investment will upgrade the Internet infrastructure which ensures the availability of the system.
4. Time
Clear role differentiation of the team avoids confusion and work overlapping, which ensures efficiency, reducing time waste. Internet infrastructure upgrades speed up the development progress, shortening the development time period. The utilization of the phased development approach breaks the whole progress into different stages, which is also beneficial to shortening the development progress.
Thus, this project is viable. The final product will optimise the management and monitoring of student information

2.0 Executive Summary

Lack of a proper administrative system in any organization makes record keeping difficult, resulting in delays and errors in the services provided by the organization. In order to solve this issue, a proper, fully functional administrative system needs to be developed and deployed. The case study chosen as part of this feasibility report is a student administration system. A student administration system is a system used by schools to perform functionalities like recording, analysing and managing student related information. In addition, this system aims to simplify information tracking for both, parents and the school's administrative staff (Staff, 2025).

The student administration system aims to monitor the performance of students, student attendance, manage discipline records and extra curricular activities. In doing so, the student administration system will boost up student administration service provided by the school, together with addressing major issues like lost student data and tracking the academic achievements of each student. To add on, this system also aims to increase efficiency and minimize administrative workload through a successful creation and operation of a structured SAS system which will manage all student related information.

The feasibility studies conducted as part of this report are economic feasibility, technical feasibility, resource and time feasibility, risk feasibility and social and legal feasibility. Economic feasibility refers to assessing how cost effective the proposed system is. Economic feasibility will ensure that the proposed system is financially viable to within the resources available (Economic Feasibility: Examples & Methods | Vaia, n.d.). Technical feasibility determines whether the proposed system can be developed and implemented with the technologies and tools that are available. Some of the aspects considered as part of technical feasibility are the hardware and software requirements, technical expertise and integrating the system being developed with a system that is already existing (Apriorit, 2024). The resource and time feasibility considers two things, firstly,

project deadlines are analysed. This involves analysing the time that will be taken by the development team to produce the final working system and the impact on the organization if the system is not completed within the deadline. Secondly, all the types of resources needed to develop the system are evaluated. For instance, technological and human resources are evaluated in the resource and time feasibility study (GeeksforGeeks, 2024). Risk feasibility is a feasibility study that is used by the system development team to assess the potential risks involved in the development of the system. Addressing risks early will help the team to save time, money and resources (What Is a Feasibility Study? 6 Types and Practical Tips, 2024). The social and legal feasibility are seen as two points of views. The social feasibility assesses the preferences and the expectations of the people that will be using the system while the legal feasibility aims to uncover the legal barriers while implementing the system, data protection and copyright (GeeksforGeeks, 2024).

Upon the completion of this system, it should ensure that data is private and secured, be able to schedule class timetables, manage library resources, track students' performance, store individual student data and provide a module for the school's staff management.

3.0 Background Information

In this modern era known as the information age, companies and organizations heavily rely on structured and well managed data for important tasks like decision making. Today, most sectors such as Transportation, Healthcare and Entertainment have adopted modern data management techniques. This, however, does not hold true for the Education sector, which has been known to, and still makes extensive use of traditional methods of data management such as paper documents and filing cabinets (NAM, 2024). While this may have worked out for schools a few decades ago, not having a dedicated academic information system today will end up costing a lot of money and time due to factors such as operational efficiency.

3.1 Existing system

The current system being utilized at the secondary school revolves around manual record keeping and documentation of information. The school principal and heads of each department currently make timetables for each class themselves and distribute physical copies at the beginning of each academic term. Class teachers are responsible for taking attendance for students using printed attendance sheets. which need to be submitted to the staff office daily. Teachers are also required to make separate marksheets per student per assessment to keep track of a students' academic performance using a spreadsheet software. Information from the school is communicated to parents through printed notices sent home with students and scheduled parent-teacher meetings held once per academic term. Enrolment for new students requires various paper forms to be pick up from the school and filled out by a parent, which is then manually entered into separate departmental records. Details about each student (such as medical information and emergency contact) is kept in a file which is stored in a cabinet in the staff office.

3.2 Stakeholder Overview

Stakeholders are people or groups that have interest in the software that is being engineered. They heavily influence or be affected by the outcomes of the proposed system (Hull et al., 2005). The 2 main types of stakeholders are internal and external stakeholders. Internal stakeholders are people within the organization that have direct interest, whereas external stakeholders are people outside the organization that have direct or indirect interest.

3.2.1 Internal Stakeholders

School Administration - This consists of the Principal, Vice Principal, Assistant Principal and board members who oversee the entire school system. They take part in strategic and tactical planning for long term goal setting and planning.

Academic Staff - This includes department heads and subject teachers who carry out the lessons, making timetables, creating assessments, taking attendance and other general tasks. They act as the bridge between the internal system and the students.
Students - This refers to the students that are currently enrolled in the secondary school and attend classes. They are required to take part in academic activities and can optionally participate in extracurricular activities such as sports and clubs.

3.2.2 External Stakeholders

Parents and Guardians - These are the primary caregivers that oversee their child's performance and provide financial, academic and emotional support. They are also responsible for filling consent forms and attending parent-teacher meetings.
Educational Authorities - This refers to the Ministry of Education which sets standards and policies that govern academic institutes. They set the overall outline of each academic year, such as deciding on external examinations and holiday dates.
Higher Education Institutions - This consists of tertiary level educational bodies such as Universities. They will require various documents from students during the enrolment process, which will need to be issued by the secondary school.

3.3 Problem Statement

The core issue that the current system faces is fragmented data management. This simply means that data is distributed across different locations (Cohesity, 2025). This problem is further amplified as the data that is being produced is tracked manually, with very little to no communication between departments. The combination of these issues has led to the rise of the following challenges:

- Difficulty in monitoring metrics related to students such as their academic performance, attendance and extracurricular records.
- Frequent delays and errors that prevent timely and accurate decision-making due to lack of communication between departments.
- Ineffective communication between stakeholders such as students, instructors, and parents, which leads to misunderstandings.
- Lack of a dedicated system prevents aggregation and analysis of larger volumes of data which prevents informed decision-making.
- Inefficient enrolment processes and schedule coordination due to data fragmentation, causing operational conflicts.
- Increased risk of misplacing or even losing important student related data as a result of manual record keeping.

4.0 Elicitation Techniques Applied

Requirements elicitation is the process of determining the requirements of the new system from core stakeholders (GeeksforGeeks, 2024). To gather requirements for the proposed student administrative system, the development team used three primary techniques. These consists of documentation, interviews and ethnography. These techniques helped get a better understanding of the users' needs from various different perspectives.

4.1 Documentation

Information that had been documented by the secondary school was acquired and analysed. This included core business items such as attendance sheets, student report cards, student registration forms, academic transcripts, class timetables, newsletters and consent forms, as well as administrative documents such as staff handbooks, library records and event plans. This helped the development team determine the important data structures that will be present in the new system and identified areas where data fragmentation was dominant.

4.2 Interviews

Multiple interviews were conducted with core stakeholders using a hybrid approach, where the interview starts off as a closed interview and transitions into an open interview after pre-determined questions had been asked. Stakeholders that were interviewed include the school principal, the vice principal, the assistant principal, teachers, student leaders (head boy and head girl) and parents from the school board. This process helped determine that the

stakeholders' biggest issues were and what they are expecting from the new system.

4.3 Ethnography

Ethnographic observations were made on core stakeholders to understand how they worked in their natural environment. This consisted of observing teachers taking attendance and submitting attendance sheets, parent-teacher meetings, the student report generation process, head of departments creating timetables and administrative staff distributing enrolment forms. This provided the development team with real world context of how the current system works and helped identify some stakeholder needs that were not obvious from the previous techniques.

5.0 Proposed System

5.1 Description of the System

The Student Administration System (SAS) is a centralized digital platform designed to address the fragmented data management, manual record-keeping, and lack of connection between departments in schools. It aims to streamline student administration, enhance record-keeping, and foster better communication among students, educators, and parents while ensuring data security and compliance with privacy standards.

5.1.1 System Objectives

- Develop and implement a system which is able to collect, store and manage comprehensive student information.
- Enhance operational efficiency and reduce administrative workload and services through the use of automation.
- Facilitate clear and streamlined communication channel between students, parents and educators through the platform.
- Simplify the student enrolment, class scheduling and academic performance tracking processes.
- Establish a centralized database that provides real-time access to student records, attendance, and participation in extracurricular activities
- Maintain data security and comply with student privacy regulations and standards

5.1.2 Core Functionality

1. **Student Information Repository**
    - Personal details.

    - Enrolment details and academic performance history.
    - Medical records and emergency contacts.
2. **Academic Performance Tracker**
    - Report card generation system
    - Assessment and grading management
    - Performance analytics and trends
3. **Discipline and Attendance Management**
    - Daily attendance tracking
    - Absence management and reporting
    - Discipline incident recording
4. **Class Timetables and Scheduling**
    - Schedule conflicts resolution.
    - Calendar integration and notifications.
5. **Extracurricular and Activities Tracker**
    - Sports and activity enrolment.
    - Document extracurricular and activity participation.
    - Student retention analytics.
6. **Communication Platform**
    - Parent-teacher communication portal.
    - Announcement and notification system.
    - Document sharing capabilities.
7. **Staff Management Module**
    - Staff profiles and qualifications.
    - Workload schedule.
    - Professional evalucation records.
8. **Library and Learning Resources**
    - Resource catalogue and availability
    - Borrowing and returning system
    - Digital resource access
9. **Data Privacy and Security**
    - Role-based access controls
    - Data encryption and protection
    - Backup and recovery systems

5.1.3 Data Capturing

- **Student Information Input:** User-friendly interfaces for administrators, teachers, and staff to input student data.
- **Attendance Tracking:** Integration with existing attendance systems or manual input through web/mobile interfaces.
- **Academic Records:** Teachers will input grades, assignments, and other academic data directly into the system.
- **Discipline Records:** Documentation of behavioural incidents, interventions, and outcomes.

- **Extracurricular Activities:** Tracking student enrolment and participation dedicated to various school activities.

5.1.4 Data Processing

The data captured will be processed and stored on a server, which could be hosted on the cloud or on-premises:

- The system will use a Database Management System (DBMS) to store and manage student data.
- The processed data will enable features like report generation, student progress tracking, and communication.
- Analytics capabilities will provide insights into student performance, attendance patterns, and behavioural trends.

5.1.5 User Interface and Access

The system will be accessible through web browsers on desktops and mobile devices:

- The interface will be designed to be user-friendly for different stakeholders:
    - Administrators
    - Teachers
    - Students
    - parents
- The system will have different portals or dashboards for various user roles:
    - **Administrator Portal:**
        - Manage student records
        - Manage user accounts and permissions
        - Oversee staff management
    - **Teacher Portal:**
        - Record attendance and discipline issues
        - Input grades and assignments
        - Communicate with students and parents
        - View student information
        - Access timetables and schedules
    - **Student/Parent Portal:**
        - View grades and assignments
        - Track attendance records
        - Access disciplinary information
        - View extracurricular activities
        - Communicate with teachers
        - Access library and learning resources

5.1.6 Data Security

The system will employ robust security measures to protect student data:

- Use of HTTPS for secure data transmission via the use of an SSL (Secure Sockets Layer) certificate. This enables encryption of user data from user's device to client's server. This works by providing an SSL certifcate with a public key whenever a user attempts to connects to the website. The user's device and the server use this key to create a secure connection. After doing so a session key is given which encrypts the data from breaches (Cloudflare, 2025).
- Access control policies to restrict unauthorized access.
- Multi-factor authentication for enhanced security.
- Compliance with student privacy regulations and standards.

5.2 Advantages and Disadvantages of the Proposed System

5.2.1 Advantages

- Improved efficiency in student record-keeping and reduction in paperwork
- Enhanced communication among school stakeholders (teachers, students, parents)
- Better tracking of student progress, attendance, discipline, and extracurricular activities
- Data-driven decision-making through reporting and analytics
- Increased data security and compliance with privacy regulations
- Centralized database allowing quick access and retrieval of student information
- Real-time tracking of academic performance and behavioural patterns
- Streamlined enrolment and scheduling processes
- Improved resource management for library and learning materials

5.2.2 Disadvantages

- Potential for high initial setup and implementation costs.
- Requirement for staff training on the new system.
- Dependence on technology and potential disruptions due to downtime.
- Potential resistance to change from users.
- Need for continuous maintenance.

5.3 Staffing

5.3.1 Skill Matrix

Skills	Rohan Nandan	Anav Chand	Vishant Kumar	Zhixian Chen	Aryan Singh
Requirements Elicitation	2	3	5	2	4
System Architecture	4	4	2	4	5
Backend Development and Management	4	5	3	3	3

UI/UX Design	3	5	5	5	2
Quality Assurance Engineering	3	5	3	2	3
System Integration	5	3	0	3	0
Software Development	2	3	3	3	3
Documentation	5	2	4	5	5
Project Management	3	2	4	2	5
Data Security	3	1	5	2	4
Total	34	33	34	31	34

Legend and Key:

Proficiency Level	Score
Novice	0
Beginner	1
Intermediate	2
Proficient	3
Advanced	4
Expert	5

5.3.2 Activity Matrix

The activity matrix below showcases the shows the tasks needed to accomplish the project, the responsible party and the overall timeline for each activity as well as the accountable head for overseeing each activity.

#	Activity	Description	Responsible Party(s)	Accountable	Consulted	Informaed	Timeline
1	Requirements Gathering	Collecting detailed requirements from school departments regarding student data management needs	System Analyst	Project Manager	School Principal, Department Heads, Teachers	All stakeholders	Week 1-2
2	System Architecture Design	Analyzing requirements and designing	System Architect	Project Manager	System Analyst, Database	Department Heads	Weeks 3-4

		database structure, user interfaces, and system architecture			Specialist		
3	Backend Development	Creating database schema, relationships, and implementing data structure for student records	Database Developer	System Architect	System Analyst	IT Department	Weeks 5-6
4	Frontend (UI/UX) Design	Designing user interfaces for dashboards, attendance tracking, grade reporting, and parent portals	UI/UX Designer	Project Manager	Teachers, Admin Staff	School Principal	Weeks 7-8
5	Backend Functationality Expansion	Designing user interfaces for dashboards, attendance tracking, grade reporting, and parent portals	Backend Developer	System Architect	Database Developer	Project Manager	Weeks 9-10

6	System Integration	Connecting all modules into a cohesive system with consolidated data flow	Integration Specialist	System Architect	All Developers	Project Manager	Weeks 11-12
7	Testing and Quality Assurance	Conducting comprehensive testing including unit, integration, system, and user acceptance testing	QA Engineer	Project Manager	Developers, End Users	School Principal	Weeks 13-14
8	Documentation	Creating technical documentation, user manuals, and training materials	Technical Writer	Project Manager	System Architect, Developers	IT Department	Weeks 15-16
9	Training	Conducting training sessions for administrators, teachers, and support staff	Training Coordinator	Project Manager	Technical Writer	End Users	Weeks 17-18
10	Data Migration	Transferring existing student	Database Developer	System Architect	Admin Staff	Department Heads	Weeks 19-20

		records from legacy systems/paper records to new SAS					
11	Deployment	Full system implementation and go-live activities	Project Manager	School Principal	IT Department	All Stakeholders	Weeks 21-22
12	Maintenance & Support	Ongoing technical support, bug fixes, and system updates	IT Support Team	IT Administrator	-	End Users	Ongoing

5.4 Space Requirements

5.4.1 Cloud Space Requirements

- The SAS will be deployed on AWS EC2 (Amazon Web Services Elastic Cloud Compute) infrastructure, elimiating need for on-premise server rooms.
- Server Specifications:
  - 16 virtual CPUs
  - 32GB ram
  - 400GB storage

5.4.2 Physical Space Requirements

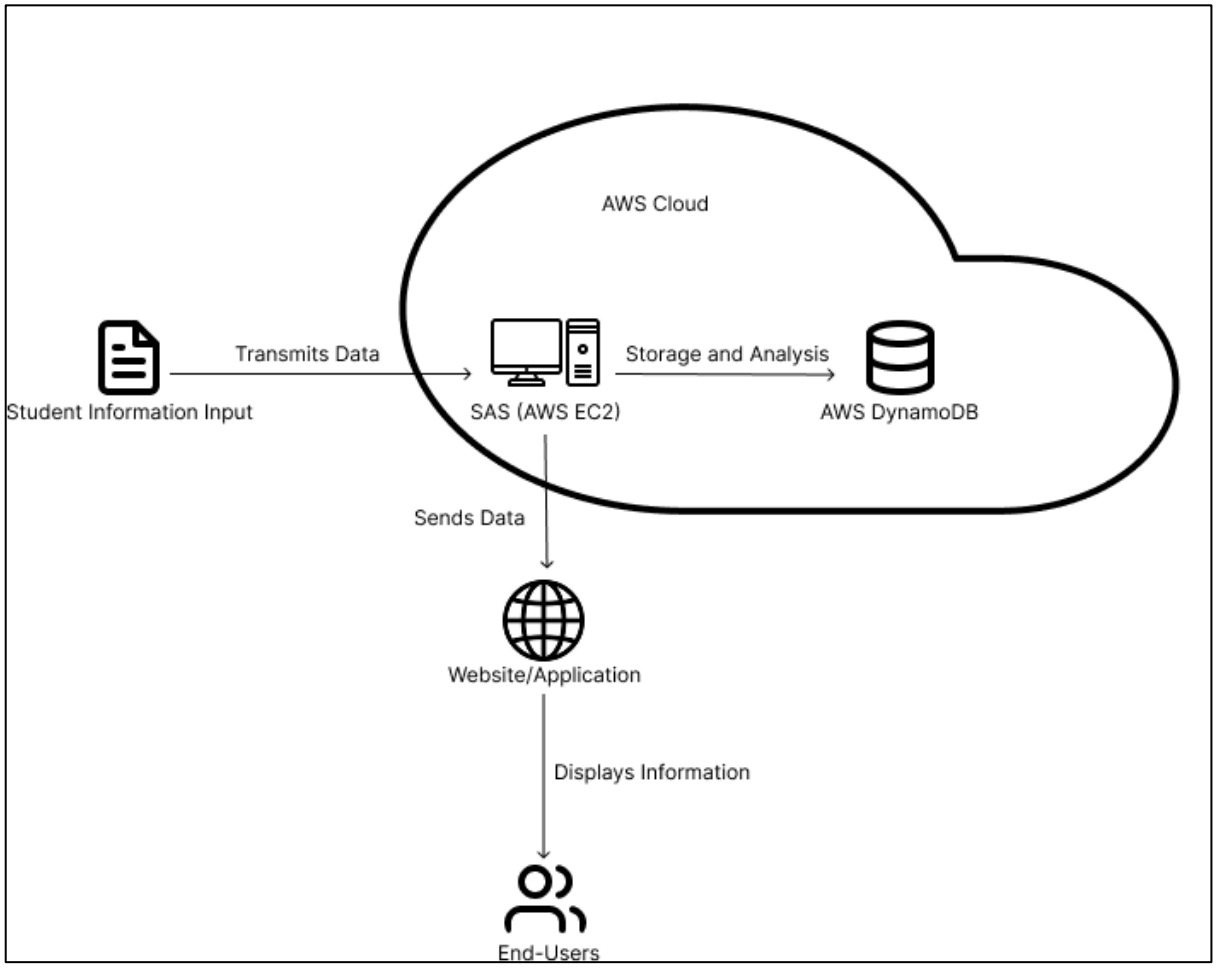
Despite the SAS being hosted on the cloud, physical infrastructure to access the software will be needed:

1. **Network Infrastructure**:
   a. Reliable Wi-Fi coverage in libraries.
   b. Sufficient bandwidth to support concurrent system access.
2. **Administrative Workstations**:
   a. Dedicated computers for administrative staff.
3. **Teacher Access Points**:
   a. Classroom computers or tablets for attendance taking and grade entry.

5.5 Basic Layout: System Architecture

The Student Administration System (SAS) follows a modular, scalable architecture designed to support all core functions while ensuring security, performance, and reliability.

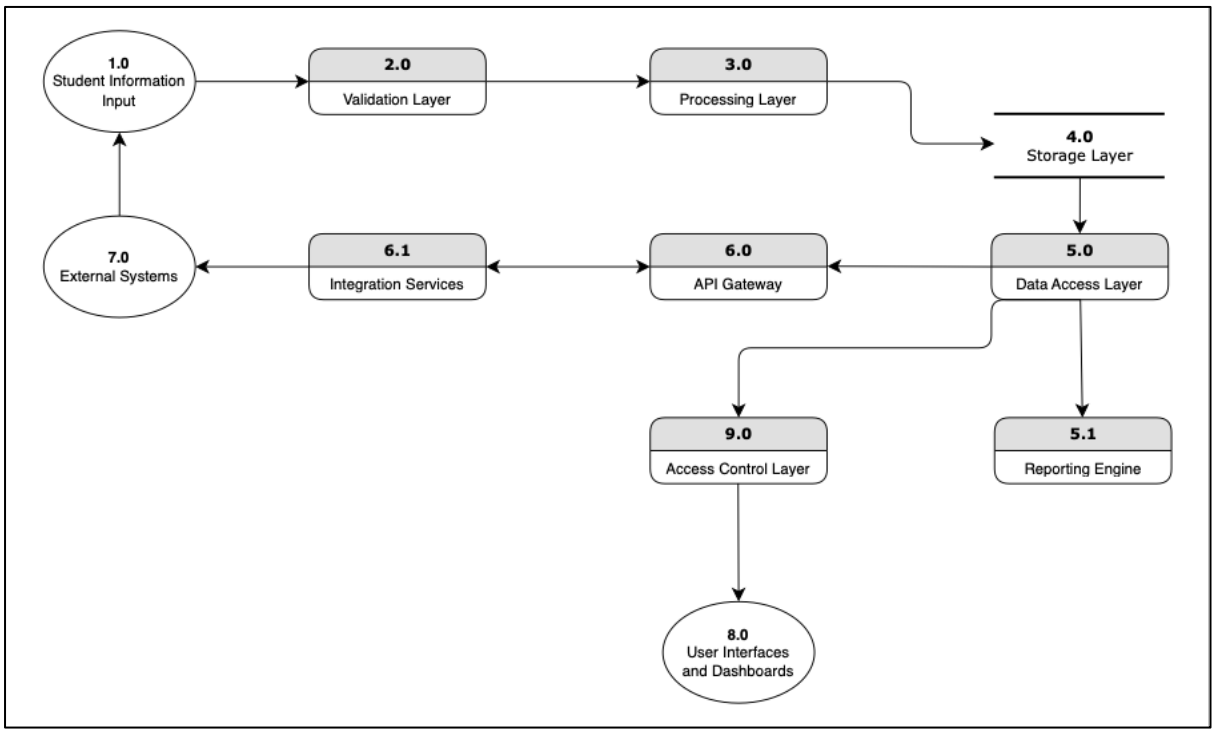
5.5.1 System Architecture



5.5.2 Data Flow Architecture

The data flow architeture displays the way data is passed into the system.

5.5.2.1 Data Flow Diagram



5.5.2.2 Data Flow Diagram Detailed Overview

The diagram starts at student information input, whereby a user is able to input information into the system. This serves as the entry point for all student-related data. Information can also import data such as from forms and records.

Once the data is received, it goes into the validation layer, where it undergoes formating and business rule validation to ensure integrity. This ensures all the information entered is accurate and of the required data type. The validated data then moves to the processing layer to execute workflow operations and apply the business logic.

The data is turned into the storage layer to manage database operations and data storage structures. Following this is the data access layer which controls all data retrieval and manipulation requests. This layer branches off into two sides. Next is reporting engine which does advanced queries to produce reports, analytics and data diagrams.

An API gateway is used to ensure efficiency between the internal and external components. Moreover, this acts as a central access point for all data services. Adding to this is the Integration services process, which handles communication protocols with the external systems. Through the external systems and integrations process, third-party systems are able to be included on the system and extend its functationality beyond internal boundaries.

The penultimate process is the access control layer which manages security and user autentication for different roles such as administrator, teacher and student. Moreover, this layer ensures certain features and data are exclusive to specific users types. After this is the user interfaces and dashboards, where end-users

are able to view the contents of the SAS. The data visualizations and analysis from the reporting engine layer are also shown here to the users.
To conclude, this form of data architecture acts to provide a logical and secure flow of data in the Student Administrative System and ensures data will always be accurate and can be used to make enriching analysis.

5.5.3 Security Architecture

- **Perimeter Security:** Firewalls, intrusion detection/prevention systems
- **Application Security:** Input validation, output encoding, session management
- **Data Security:** Encryption at rest and in transit, data masking for sensitive information.
- **Identity Management:** Role-based access control, multi-factor authentication
- **Audit Trail:** Comprehensive logging of all system access and changes

5.5.4 Technology Stack

- **Frontend:** HTML5, CSS3, JavaScript frameworks React
- **Backend:** Python/Django - Cross Site Request Forgery (CSRF) and Node.js
- **Database:** PostgreSQL and Amazon Web Services DynamoDB
- **API Management:** RESTful services
- **Security Protocols:** SSL encryption, multi-factor authentication, role-based access control.
- **User Interface:** Web-based dashboard accessible via desktop and mobile
- **Reporting Tools:** Reporting engine for generating academic reports, attendance summaries, and analytics.
- **Communication Tools:** In-built messaging capabilities between stakeholders via forums.

5.6 Cost-Benefit Analysis

Assuming:

- 1000 concurrent students
- 10 Staff
- 2 Administrators

Cost Factors	**Estimated Cost**	**Benefits**
Software Development	High (Initial)	Improved efficiency, reduced errors, elimination of manual paperwork
Hardware/IT Infrastructure	Medium	Enhanced data management, secure storage and processing
Training	Medium	Effective system utilization, ensures proper adoption

Maintenance	Ongoing	System reliability, data security, long-term accuracy
Module Integration	Medium	Seamless operation across all school departments
Security Implementation	Medium-High	Protection of sensitive student data, compliance with regulations

5.6.1 Fixed Costs

As found in 5.6, software development, hardware/it infrastructure implementation costs will be the most prevalent and demanding fixed cost. It was found that the cost to creae a system which would be able to have able to handle 1000 concurrent users would be $90,000 (ScienceSoft, n.d.). Moreover, the costs involved in setting up workstations for the administrators, teachers and computer laboratories would be approximately $30,000, accounting for each workstation costing $1400 (Penn Information Systems and Computing, 2025). Lastly, the costs to implement the new system into the high school ecosystem would be $8,000 (Charest, 2023).

Title	Description	Cost
Software Development	The actual software development team who will be working on the project. Scheduled to be between 10 to 12 weeks with the team working fulltime.	$80000
Hardware/IT Infrastructure	• Networking equipment • Databases • Workstations for staff • Backup systems • Cloud service setup	$30000
Implementation Cost	• Data migration from existing systems • System configuration • Installation • Testing the implemented system	$8000

5.6.2 Ongoing Costs

Ongoing costs are recurring expenses necessary to keep the system operational. Maintenance includes things like system updates, bug fixes, technical support, and potentially server costs.

Title	Description	Cost

Maintenance and Support	• Software updates and patches • Technical support (in-house or external) • Database administration • Server maintenance (if on-premises) • Cloud service fees (if cloud-based)	$15,000
Training	• Developing training materials • Conducting training sessions • Trainers' time	$8,000
Operational Costs	• Account for any increased operational expenses. • Electricity consumption • Additional IT staff (if needed) • AWS EC2 and DynamoDB	$17,000
System Administrator Services	• Moderation for system • Support for updates	$40,000

5.6.3 Benefits of the System

5.6.3.1 Tangible Benefits

- Reduced Paperwork Costs.
  - Estimate savings in paper, printing, and storage.
  - Reduction in staff time spent on manual tasks.
- Increased Efficiency:
  - Time saved in administrative tasks (e.g., enrolment, scheduling).
- Reduced Errors:
  - Cost saved in correcting errors in the old system.
  - Minimizing errors with the new system.

5.6.3.2 Intangible Benefits

- Improved Communication:
  - Enhanced communication between teachers, students, and parents.
  - Better information flow within the school

- Enhanced Data Management:
    - Centralized database for easy access to information
    - Improved data accuracy and reliability
- Better Decision-Making:
    - Data-driven insights for informed decisions
    - Improved reporting and analytics
- Increased Security:
    - Protection of sensitive student data
    - Compliance with privacy regulations
- Streamlined Processes:
    - Efficient enrolment and scheduling
    - Simplified access to learning resources

5.6.3.3 Annual Benefits Calculation

Description	Contents	Estimated Benefit
Administrative Task Automation	Attendance Grading Scheduling	$35,000
Reduced Errors	Cost saved in correcting erros.	$20,000
Reduced Paperwork	Printing, paper and physical accomodation	$35,000
Total	Estimated capital saved	$90,000

5.6.4 Cost/benefit Analysis of the System

Fixed Costs = $80,000 + $30,000 + $8,000 = $118,000
Ongoing Costs = $40,000 + $17,000 + $15,000 + $8,000 = $80,000
Total Costs = Fixed Costs + Ongoing Costs = $198,000

Total Evalucation = Total Benefits – Total Costs = **-$108,000**

While the initial installation of this system will incure a loss and will require significant time and effort to management, the non-monetary gain from the system would be substancial for the high school. The system will enable management to effectively monitor and manage student progress and performance. This will lead to the system to ultimately save money and recoup for its costs. This new system would be a worthwhile and cost-effective investment in the dissemination of information and resources within the high school environment

5.7 Computer Software Requirements

5.7.1 Desktop Software Requirements

- **Operating System:** Windows/Linux-based

- **Accessibility**: Web browser

5.7.2 Mobile Software Requirements

- **Operating System:** Android/IOS
- **Accessibility**: Web browser or mobile application

5.8 Site Possibilities

- **Cloud Deployment:**
  - Best for scalability, remote access, and minimal hardware requirements.
  - SaaS model with browser-based access.
  - Elastic scaling to accommodate peak usage periods
  - Automated backup and disaster recovery
  - Reduced on-premises infrastructure requirements
  - Continuous updates and maintenance
- **Mobile Application:**
  - Extended accessibility through dedicated apps for smartphones and tablets
  - Native applications for iOS and Android devices
  - Offline capabilities for key functions (attendance, grades review)
  - Push notifications for alerts and updates
  - Mobile-optimized APIs for efficient data transfer
  - Secure local storage with encryption for offline data
  - Simplified workflows designed for smaller screens and touch interfaces

6.0 Feasibility Study

In order to determine whether the proposed SAS system is feasible or not, by using the technical, operational, financial and the legal feasibility, and whether the feasibilities align with the system goals and objectives or not, a feasibility study is a methodical and well-documented analysis that is carried out early during a project (GeeksforGeeks, 2024). It also helps organizations and stakeholders to evaluate and understand associated risks and help determine

the return on investment (ROI) on the system being built (Team, 2025), and it is also a way in which companies might measure success (TheKnowledgeAcademy, n.d.), the potential benefits, and the costs associated with the project development to decide whether a project should be given the green light, modified, or simply abandoned (What Is a Feasibility Study? 6 Types and Practical Tips, 2024).

6.1 Economic Feasibility

The economic feasibility analysis is done to evaluate whether the School Administration System (SAS) is financially sustainable by comparing the costs incurred during development and operations to the potential benefits that the new system will offer (Team, 2025). The factors that influence economic feasibility are:

6.1.1 Development Expenses

6.1.1.1 Software

The SAS system leverages the use of **open-source frameworks** such as Django for backend and React (JavaScript Library) for frontend development, PostgreSQL for databases, and DynamoDB for database management to eliminate the need to pay licensing fees, which reduces upfront costs. Secure access to updates and troubleshooting materials are ensured by the widespread adoption and recognition of such frameworks amongst developer groups, development platforms like GitHub (collaborative version control and management) and IDEs such as Visual Studio Code (code editing) all assist in facilitating teamwork. This approach emphasizes the concept of flexibility and cost-efficiency while at the same time also maintaining modern functionality and quality standards.

6.1.1.2 Hardware

The SAS system model is **cloud-hosted on AWS EC2** (Amazon's Dynamic Site Hosting Platform, requirements: 16 virtual CPUs, 32GB RAM, and 400GB of storage capacity) to avoid physical server costs. The cloud-hosting approach offers scalability, robust security, and reliability, which also ensures that the system is fully capable of handling peak loads (use case example: report card generation). End users only require basic devices (i.e., i3 processors, desktops with as low as 4GB RAM, or Android/iOS smartphones) and security; this makes the system accessible to schools and the families of students with limited resources.

6.1.1.3 Training

School Teachers and staff can attend workshops to learn system features such as attendance tracking, gradebook management, automation of report cards, and user guides for navigation through the system portals (*School Administration*, n.d.). User training will ensure smooth adoption, and feedback

tools during the training will help refine the usability of the system (based on user input experience).

6.1.1.4 Deployment

A pilot approach during deployment (trying out the new system in only one part of the organization) e.g., testing in Year 13 only would minimize risk levels to the organization by identifying issues early. Gradual integration (e.g., replacing attendance registers first and using the system's attendance tracking feature) would allow users to adapt incrementally to the new system, which in turn would reduce disruption to daily organizational workflow.

6.1.1.5 Maintenance

Annually scheduled updates for the system would address system bugs, implement security patches, and feature enhancements (e.g., new report card templates). The maintenance aspect can be handled by a part-time IT administrator (cost: $40,000 annually, source: Google) to handle troubleshooting and ensure system reliability without incurring full-time staffing costs.

6.1.2 Operational Costs

Description	*Defined*	*Cost*
AWS EC2 & DynamoDB	Cloud services for server-hosting and database management	$12,000
System Administrator	Part-time system support for updates	$40,000
Internet/Electricity	Utility costs for system usage	$5,000
System Maintenance/Updates	Software improvements/Security patches	$15,000
User Training	Annual workshops/videos for new staff/parents	$8,000
Total Cost	Cost to maintain system performance	$80,000

Key Take: The total costs total up to $80,000; this value reflects the essential investments required to maintain the system performance and ensure user satisfaction.

6.1.3 Annual Benefit Estimation

Description	*Defined*	*Estimated Benefit*
Reduced Administrative Hours	Task Automation (attendance, grading)	$35,000
Fewer Errors in Records	Digital records minimize manual entry	$10,000
Improved Parental Engagement	Real-time communication boosts enrolment retention (fewer school switches)	$20,000

Data-Driven Choices	Analytics enable targeted intervention (tutoring and assistance for underperforming students)	$25,000
Total	Estimated capital saved	$90,000

Key Take: The total expenses come to $80,000, which represents the expenditures required to keep the system operating smoothly and guarantee user satisfaction. The table above shows that the benefits outweigh the expenses by $10,000 in a single year and also offering other advantages such as efficient system performance and enhanced school reputation.

6.1.4 Conclusion

It can be said that the proposed SAS system is economically viable, with an expected net annual benefit of $10,000 (benefit vs. costs). The long-term advantages include scalability (expansion to more grades), reduced paperwork, and stronger parent-school collaboration. A phased approach to the system rollout (beginning with attendance tracking) will help mitigate upfront risks, whereas cloud hosting ensures adaptability for future needs. The efficiency gains overtime and error reduction help increase and amplify financial returns, which will help maintain the sustainability of the system.

6.2 Technical Feasibility

The technical feasibility analysis is done to determine whether the proposed SAS system can be developed and put into operations effectively using the in-house resources (TheKnowledgeAcademy, n.d.), that is, the infrastructure and technology that are now accessible, and it also assists companies to evaluate the technical expertise of the development team to ensure the successful creation of the suggested system (GeeksforGeeks, 2024).

6.2.1 Required Software

6.2.1.1 - Frontend Software

A **JavaScript framework** called **React.js** is used by the SAS system to create a dynamic and responsive online interface that is able to accommodate various screen sizes. The component-based architecture of React.js allows for **reusable user interface elements** (e.g., attendance forms and automated gradebooks), which will help speed up the development phase. Parents with limited resources can also use their daily mobile devices and tablets to access the mobile app, thus ensuring inclusivity for all device peripherals.

6.2.1.2 - Backend Software

The SAS system uses **Node.js** (JavaScript runtime) and **Django** (Python framework) to handle all **server-side logic operations** (user authentication and

data processing). Django's built-in security features (such as Cross Site Request Forgery (CSRF) protection) and Node.js's scalability features make them the ideal choice for constructing REST APIs (methods of communication between the front and backend systems) to connect the frontend and the system database.

6.2.1.3 - Database Systems

The SAS system uses the **PostgreSQL** database system for structured data such as student records, grades, and school timetables. It also ensures data integrity and provides query support. It also uses **AWS DynamoDB** (a cloud storage database) for unstructured and semi-structured data (user activity and logs). It has a serverless architecture which can scale automatically with user demand, which will reduce wait time during peak usage (events such as report card generation)

6.2.2 Required Hardware

6.2.2.1 - Cloud Infrastructure

The use of **AWS EC2 virtual servers** helps supply the computing power (16 CPUs, 32GB RAM) to host the SAS backend operations and handle all concurrent users (e.g., different teachers simultaneously updating student grades). Also, the use of **AWS S3 servers** for having a secure and scalable storage for user backups (e.g., student records and data recovery) in the event of system failure or downtime.

6.2.2.2 – User Devices

The school is already equipped with basic desktops (in the staff room, computer lab, or library) with system specifications as low as Intel i3 processors and 4GB RAM, which are sufficient for accessing the SAS system that is being developed.

6.2.3 User Interface/User Experience

6.2.3.1 – Accessibility

Use of **semantic HTML5 code** ensures compatibility with screen readers and resolution (e.g., for the visually impaired) and keyboard navigation. Also, the use of **colour contrast in compliance** with WHATWG standards (e.g., text-to-background contrast ratios) to accommodate for users with colour blindness issues.

6.2.3.2 – Responsive Layout

Having a **simplified menu and collapsible sections** in the system for smaller screens, thus allowing **mobile optimization** (e.g., parents checking their child's

attendances on their personal devices), and for desktops, there can be fully featured interfaces with full functionality, mostly handy for administrative tasks (e.g., generating student report cards)

6.2.3.3 – Navigation Workflow

Use of the **3-click rule,** where the user should be able to get the desired choice within 3 clicks, and thus promote a streamlined and organized workflow with straightforward processes (e.g., for entering student test marks: selecting class → selecting student → entering mark), which helps reduce user frustration and chances of errors. Users can easily get what they want.

6.2.4 Privacy & Security

6.2.4.1 – AWS Security Measures

Data encryption using AES-256 (Advanced Encryption Standard, 256 bits), an encryption algorithm used to store sensitive data where it scrambles information into unreadable format which will require a 256-bit-long unique key to decrypt, which ensures secure information transfer and storage. AWS is also certified with compliance standards such as SOC 2; therefore, it provides reliability and infrastructure security, system scanning features, and patch updates.

6.2.4.2 – Regulatory Compliance

Student data such as grades are kept anonymous or only shared with students/parents, which adheres to GDPR regulations and avoids legal issues. other features, such as Role-Based Access Control (RBAC) between the administrators, teachers and students, are all tiered by the permissions allowed (e.g., student cannot change or modify any system settings)

6.2.5 Scalability

AWS EC2 allows **automatic scaling** (e.g., adding more servers during peak use hours, such as enrolment timeframes) to cater for increased user traffic without any downtime. Also, the use of **modular architecture** and **microservices** (such as having a separate module for attendance and another for grading) allows teaching staff to easily update the desired module without affecting the other modules.

6.2.6 Testing & Quality Assurance

6.2.6.1 - Unit Testing

The use of automated scripts to validate all individual components (niche features such as ensuring that the attendance is correctly summed up for each week and term), which are done using tools such as React and Django.

6.2.6.2 – User Acceptance Testing (UAT)

Conducting UAT using real-world scenarios (e.g., parents trying to access simulated report cards using low-end devices to test the feature functionality), then collecting user feedback to help refine usability features of the system.

6.2.7 Conclusion

Under technical feasibility, by leveraging open-source tools, using modern frameworks, and using proven and accredited technologies, and also using scalable cloud architecture together with the user-centered design methodology, the proposed SAS system is able to meet the current user needs while also having an eye toward future demands such as adding AI-driven analytic tools into the system. The phased approach for testing (Unit Testing → UAT) further helps reduce the technical risks involved, which ensures system reliability (What Is a Feasibility Study? 6 Types and Practical Tips, 2024).

6.3 Resource and Time Feasibility

The resource feasibility analysis process provides assessment on whether the organization that the project is being built for is able to allocate and manage these resources effectively (What Is a Feasibility Study? 6 Types and Practical Tips, 2024), it also helps developers to evaluate whether they have all the resources needed to complete the project successfully or not (GeeksforGeeks, 2024).

6.3.1 Resource Feasibility

6.3.1.1 Team Composition

The project development team has been structured in a strategic manner, using six members, to cover all aspects. They are:

Project Manager: A manager is responsible for overseeing timelines, budget tracking, and effective team coordination between all members of the development team. They help in risk mitigation, resolving conflicts, and ensuring that the product matches user expectations (quality control).

System Developers: There are **two developers** that are responsible for coding the system, system integration, and problem-solving (technical issues). Their job expertise ensures that the core functionality of the SAS system project (e.g., the software development) is built effectively.

UI Designer: There is **one UI designer** who is responsible for enhancing the user experience (UX) and the interface design to oversee and ensure that the final product is visually appealing to all end-users (teachers and students).

User Trainers: Two trainers are tasked with the creation of training materials and organizing workshops to conduct training for end-users on how to operate the new SAS system (usability).

Note: The team roles have been well-balanced to cover all development aspects, covering the managerial, technical, creative, and educational needs of the development phase.

6.3.1.2 Internet Infrastructure

The school-wide Wi-Fi upgrade is essential to support the project demand and peripherals such as the cloud-based architecture, high user traffic, and usage of heavy data applications such as DynamoDB. This is a one-time cost of $5,000 (as also explained under section 6.1.2). This investment aims to address and provide a solution to potential constraints such as poor connectivity and long downtime. It is done to ensure reliable access during the testing and deployment phases and that the team manages the resources effectively (What Is a Feasibility Study? 6 Types and Practical Tips, 2024).
Note: The one-time Wi-Fi cost is a manageable upfront expense where recurring fees are avoided so as to avoid a potential strain on the project budget.

6.3.2 – Time Feasibility

Time feasibility analysis is done to help the development team evaluate whether their project timelines and deadlines are realistic or not (TheKnowledgeAcademy, n.d.); it also helps them to cater in the development and testing phase, as well as a certain period of time for a buffer to ensure that the team can meet project demands or how much time they will take to complete the final project (GeeksforGeeks, 2024).

6.3.2.1 System Planning

Activity Sequence: **Define Project Scope → Requirements Gathering → Finalize Designs → Allocate Resources**
This ensures that the SAS system being developed is aligned with the system needs of the school and also minimizes the occurrence of changes mid-development (What Is a Feasibility Study? 6 Types and Practical Tips, 2024).
Estimated time: Two months

6.3.2.2 System Development

Activity Sequence: **Building Product → Integrating Features → Conducting Tests**
The most resource-intensive phase, where ample time allocation allows for iterative modifications to the system and provides enhanced quality assurance.
Estimated time: Six months

6.3.2.3 Pilot Approach—Testing

Activity Sequence: **Product Deployment (e.g., only use in Year 13 for pilot testing)** Obtaining feedback from real-time usage and end users will help developers identify any bugs and overlooked usability issues. It also helps refine the system before it is fully deployed to cater to the whole school, which helps reduce the risk of a widespread system failure.

Estimated time: Two months

6.3.2.4 Full System Deployment

Activity Sequence: **System is deployed throughout the entire school system.** This method of pilot approach prevents overwhelming the users with having to learn and so quickly adapt to an entirely new system. The use of training sessions and support from trainers helps users in learning how to operate the new system.
Estimated time: Two months

6.3.2.5 Buffer Time Period

A certain amount of buffer time can be allocated for any unexpected delays and adjustments during development, such as needing **additional time for user training** should users face difficulty adapting to the new system. Also, should **critical flaws (technical bugs)** be discovered late during the testing phase, it would require reworking, therefore chewing time.
This also helps to accommodate for common risks and minor inconveniences during development and testing, which will reduce the likelihood of missing project deadlines.
Estimated time: One month

6.3.3 – Conclusion

The proposed SAS system benefits from the team resource allocation as all team members have clearly defined roles, which avoids confusion and work overlapping, while the infrastructural upgrades (internet/Wi-Fi) cater to the functional need to have connectivity in the system. A phased approach assists the development team to break down the SAS system project into multiple different stages, which makes room for adjustments to be made to the system, depending on the user feedback received during pilot testing, whereas the 1-month time buffer provides ample time for the development team to mitigate risks (pilot testing helps reduce the impact of unforeseen issues, and dedicated trainers ensure that users get along well with the functionalities of the new system) all without having to significantly extend the project timeline.

6.4 Risk Feasibility

The risk feasibility analysis helps the development team assess whether they are able to effectively manage the potential threats to the project being developed (What Is a Feasibility Study? 6 Types and Practical Tips, 2024), where risk evaluation is carried out on factors such as data breaches and system downtime, and then forming mitigation strategies such as backup plans help make the project more viable, it also ensures that the school is able to adopt to the new SAS system safely and in a sustainable manner (Team, 2025)

6.4.1 Potential Risks

6.4.1.1 Data Breaches

There is a possibility of information leak due to weak cybersecurity, which could lead to events such as identity theft, which would result in data leaks, leaked student records (e.g., leaked exam papers, student records and personal details), leading to legal action and reputational damage.

6.4.1.2 Overrunning the Budget

Unplanned additions and enhancements to the system such as last-minute feature requests such as changing or adding user interface attributes, budget bust can also be due to a **scope creep**, both of which can cause a budget inflation beyond the original planned budget of $80,000 for the system.

6.4.1.3 User Resistance

The teaching staff may likely be still used to and familiar with the previous system, therefore they may reject the new SAS system because it is still a foreign concept for the school, which may slow the adaptation process and have a resistance towards training and accepting the new system.

6.4.1.4 System Downtime

Technical failures such as server crashes or network outages during peak usage periods such as accessing grades and student report cards can disrupt communication between the school staff and possibly lead to data loss, which would cause chaos and result in associated users no longer having faith within the new system.

6.4.2 Risk Mitigation

6.4.2.1 Multi-Factor Authentication and OTP

The implementation of a multi-factor authentication system such as Google Authenticator would add extra security to the system as users would be required to verify their logins first using their password, and then an OTP (one time password) code sent to their emails to successfully log in to the SAS system. Sensitive data can also be encrypted using the AES 256 encryption algorithm.

6.4.2.2 Controlling the budget

The project manager could make the project scope management stricter with stakeholder and all higher-ups authorizing the strict management measures to prevent the chances of unnecessary additions to the system functionality and ensuring adherence to the project plan.

6.4.2.3 Resistance to Change

Gathering feedback from end-users early during the testing phase would help the development team to design the system to their needs and meet user expectations (e.g., deploying intuitive interfaces for grading). Incentives such as recognition for early adopters can also be introduced to help encourage other users to drive towards learning the system and training peers as well.

6.4.2.4 Preparations for Downtime

The use of temporary based paper alternatives (such as attendance register sheets and exam timetables) ensures that the process is continued without halt in the event of system failure. The system can also be subject to stress-testing before deploying the system during peak usage periods to test the functionality and schedule maintenance during off-peak usage times.

6.4.3 Conclusion

The project risks that have been identified are manageable due to the protective measures taken to address these issues such as implementing features like multi-factor authentication and also making the project scope strict in order to avoid making any unplanned changes, through such measures we are able to cover both technical vulnerabilities, and the human factor involved. On the other hand, the use a balanced approach to risk management such as the early engagement of teachers into the design and testing process to obtain feedback and refine the system according to those requirements helps reduce resistance and allows for a much smoother system and user transition.

6.5 Social/Legal Feasibility

6.5.1 Social Feasibility

6.5.1.1 Staff Concerns

The system users and the teaching staff may have potential concerns that the system can add more work to their daily workload at school, this can be addressed during the user training sessions by demonstrating how the system helps users to save time by using features such as automating report cards instead of having to make them manually.

6.5.1.2 Parental Concerns

The system can be configured to send text messages and notifications using the SAS system, this helps to address the concern that not all households have regular internet connectivity. This would also resolve the issue of device types because to receive a notification, users just need to be logged in to the system using any device (smartphone/tablet or laptop).

6.5.2 Legal Feasibility

6.5.2.1 Compliance

The school is handling sensitive information which is kept confidential between the school staff, students and their parents, therefore before accessing any student data such as student grade reports, parental signatures and consent forms are required to be filled in order to access student information. This allows parental control over their child's academic information while also maintaining information privacy.

6.5.2.2 Ethics

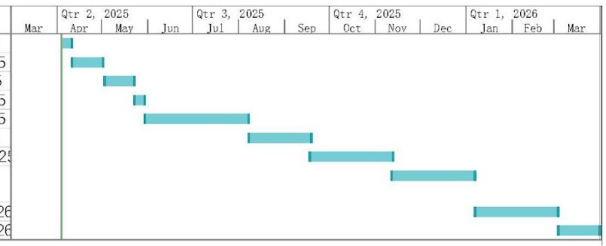
The system has in place, clear rules and uses the Role-Based-Access-Control feature (as discussed in 6.2.4.2) where the system permissions are tiered out. For instance, the student grades should only be shared to the student, their parents and their form teachers-not to the whole school staff, trust and privacy is very important when dealing with sensitive information.

6.5.3 Conclusion

The main purpose of the social feasibility study is to put the people first, by using niche features such as SMS notifications for families without internet access, we are ensuring that the SAS system helps serve the real need by keeping everyone informed and on the same page with easy-to-access information. It is important that we, as developers, ensure that there is transparency about how student data is handled and describe how every feature helps provide value to the system.

7.0 Project Schedule

ID	Task Mode	Task Name	Duration	Start
1		Define Project	1 wk	Fri 4/4/25
2		Requirement Gathering	3 wks	Fri 4/11/25
3		Finalize Designs	3 wks	Sat 5/3/25
4		Allocate Resources	1 wk	Fri 5/23/25
5		Building Product	10 wks	Fri 5/30/25
6		Integrateing Features	6 wks	Fri 8/8/25
7		Coducting Test	8 wks	Thu 9/18/2
8		Pilot Approaching Deployment (Year13	2 mons	Wed 11/12/25
9		Full Deployment	2 mons	Wed 1/7/26
10		Buffer Time Period	1 mon	Wed 3/4/26



Project: Project1
Date: Fri 4/4/25

Task
Split
Milestone
Summary
Project Summary
Inactive Task
Inactive Milestone
Inactive Summary
Manual Task
Duration-only
Manual Summary Rollup
Manual Summary
Start-only
Finish-only
External Tasks
External Milestone
Deadline
Progress
Manual Progress

Page 1

8.0 Final Recommendation

Upon completing this report, the development team recommends proceeding with the development of the student administration system. The findings of this report have provided enough evidence to call the student administration system beneficial to secondary schools. Reflecting back on the feasibility studies, the economic feasibility concludes that the proposed system is economically viable

as the benefit outweighs the cost and the system brings in advantages such as reduced paperwork. The technical feasibility concludes that the proposed system is technically feasible as it meets current user needs through the use of modern frameworks and scalable cloud-based architecture. The resource and time feasibility deems the proposed system feasible as everyone working on the project have defined roles which avoids confusion and work overlapping, ensuring that the system development is completed within the projected timeline. The risk feasibility concludes the risks involved in the project are mitigated through measures such as multi factor authentication and early user engagement. This reduces resistance towards the system and ensures a smooth transition into the new system. The social/legal feasibility concludes that there is transparency about how student related data is handled, the system also uses niche features as part of social feasibility together with providing value to the system itself. Thus, the student administration system is deemed feasible as it is declared feasible by each of the feasibility studies conducted.

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Assignment 1 Mark Allocation Sheet

After having discussed it as a group, we recommend the following mark allocation to each group member based on contribution or lack of it throughout the assignment.

Group number: 12 Project Manager: Vishant Kumar

Member Student ID	Percentage Contribution	Responsibilities
S11230430	20%	Question 2 and 8
S11221203	20%	Question 3 and 4
S11230987	20%	Question 6
S11234883	20%	Question 5
S11230686	20%	Question 1 and 7

Signed

Member Name	ID	Signature
Vishant Kumar	S11230430	vishant
Anav Chand	S11221203	anav
Aryan Singh	S11230987	aryan
Rohan Nandan	S11234883	rohan
Zhixian Chen	S11230686	zhixian